

Fabric Monitoring Made Simple

Built-In Tools and Custom Solutions

Thank you, partners



Ásgeir Gunnarsson

- **Partner at Data Lab**
 - Advisory
 - Implementation
 - Training for customers
- **Microsoft Data Platform MVP**
- **Organizer of**
 - Data Platform Next Step, Power BI Next Step and Power BI Cruise

 [linkedin.com/in/asgeirgun/](https://www.linkedin.com/in/asgeirgun/)



Just Blindbæk

- **Principal Architect at twoday**
 - Pre-sales, workshops, events, marketing
 - Internal practice development
 - Academy: External training
- **Microsoft Data Platform MVP**
- **Found and organizer of**
 - Danish Microsoft BI Community (MsBIP.dk)
 - Power BI UG Denmark (PowerBI.dk)
 - Data Platform Next Step, Power BI Next Step and Power BI Cruise

 [linkedin.com/in/blindbaek/](https://www.linkedin.com/in/blindbaek/)

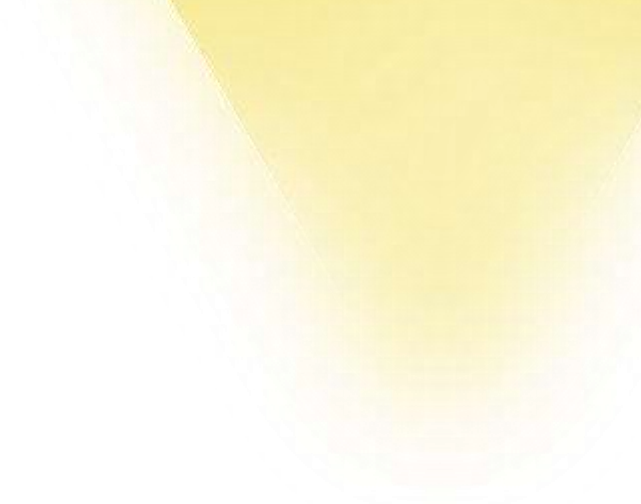


Slides



Agenda

1. Why monitor?
2. Out of the box solutions
3. Community tools
4. Build your own solution
5. Wrap-up



Why monitor?

Overview of Monitoring in Fabric

- **Operational efficiency:**

- Track performance of reports, queries, compute usage, and pipelines.
- Monitor system health and ensure uptime.
- Adoption of Fabric

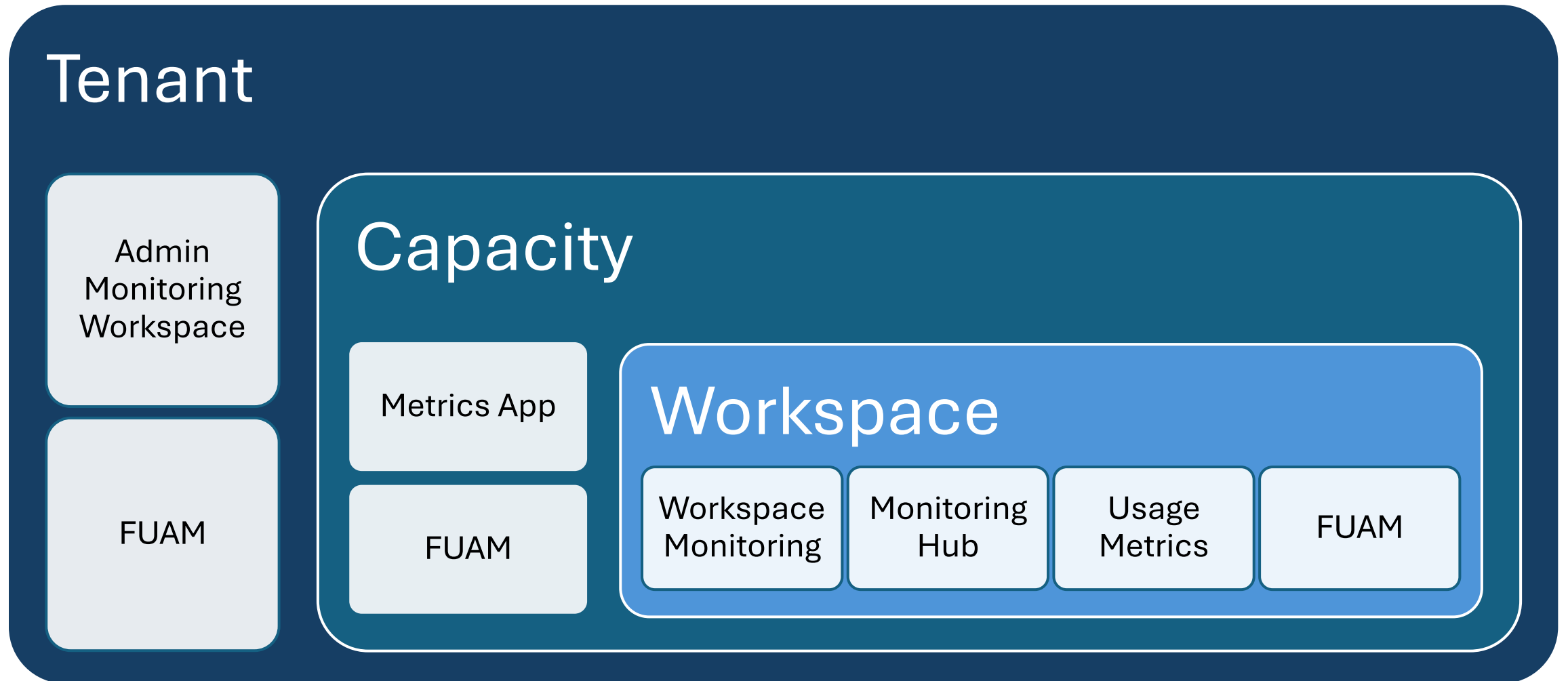
- **Security and Compliance:**

- Audit access, roles, and permissions.
- Monitor security events for regulatory compliance (GDPR, SOC).

- **Cost:**

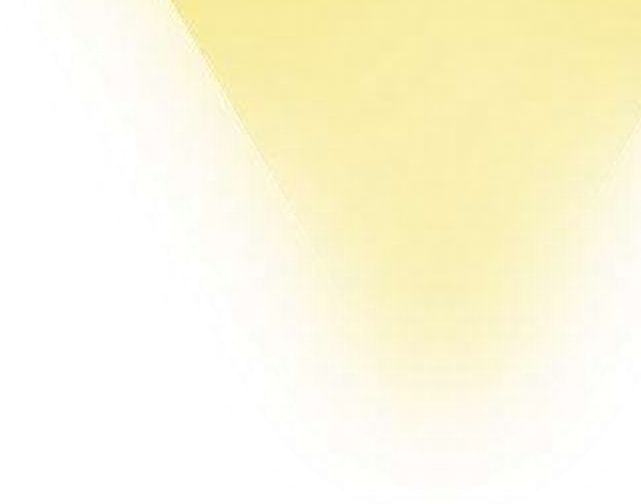
- Monitor workload usage and optimize resources.
- Set alerts for cost overruns and resource efficiency.

Scope & Tools



Out of the box solutions

1. Usage Metrics Report
2. Monitoring (hub)
3. Workspace Monitoring
4. Capacity Metrics App
5. Admin Monitoring Workspace

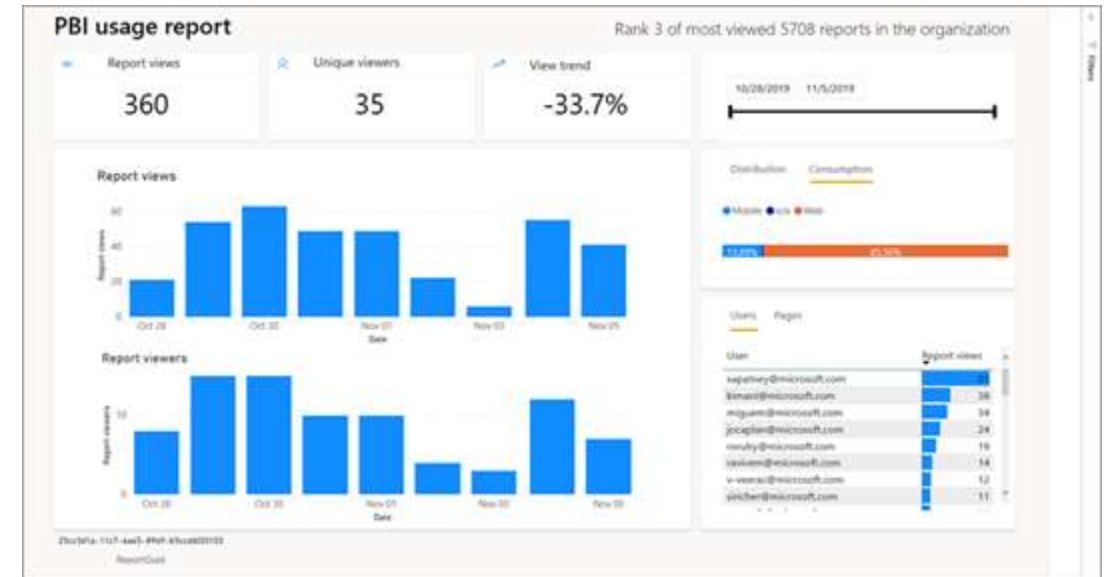


Usage Metrics Report

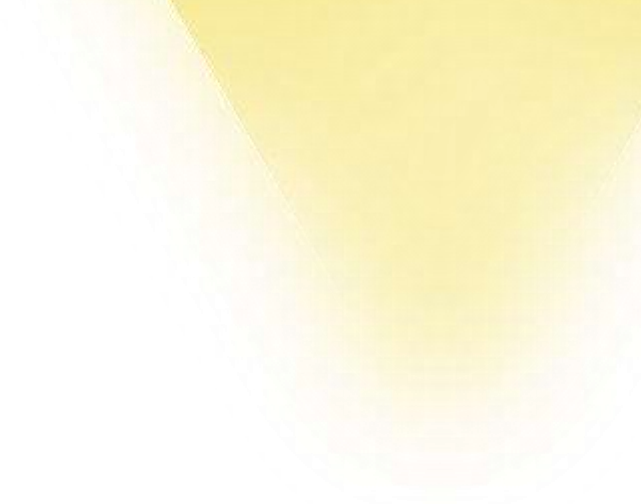
See how your content is really being used

Usage Metrics Report: See how your content is really being used

- Pre-built usage report for reports & dashboards in workspaces - with improved metrics and slicers.
- Shows last 30 days of usage by default.
- Allows slicing by platform (web / mobile), report page, and how users accessed the content.
- Built on a semantic model that refreshes daily; cannot change refresh schedule.
- Access requires: Pro or Premium Per User (PPU) license; edit permissions on the content; usage metrics enabled in tenant by admin.



<https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-usage-metrics> and
<https://learn.microsoft.com/en-us/power-bi/collaborate-share/service-modern-usage-metrics>

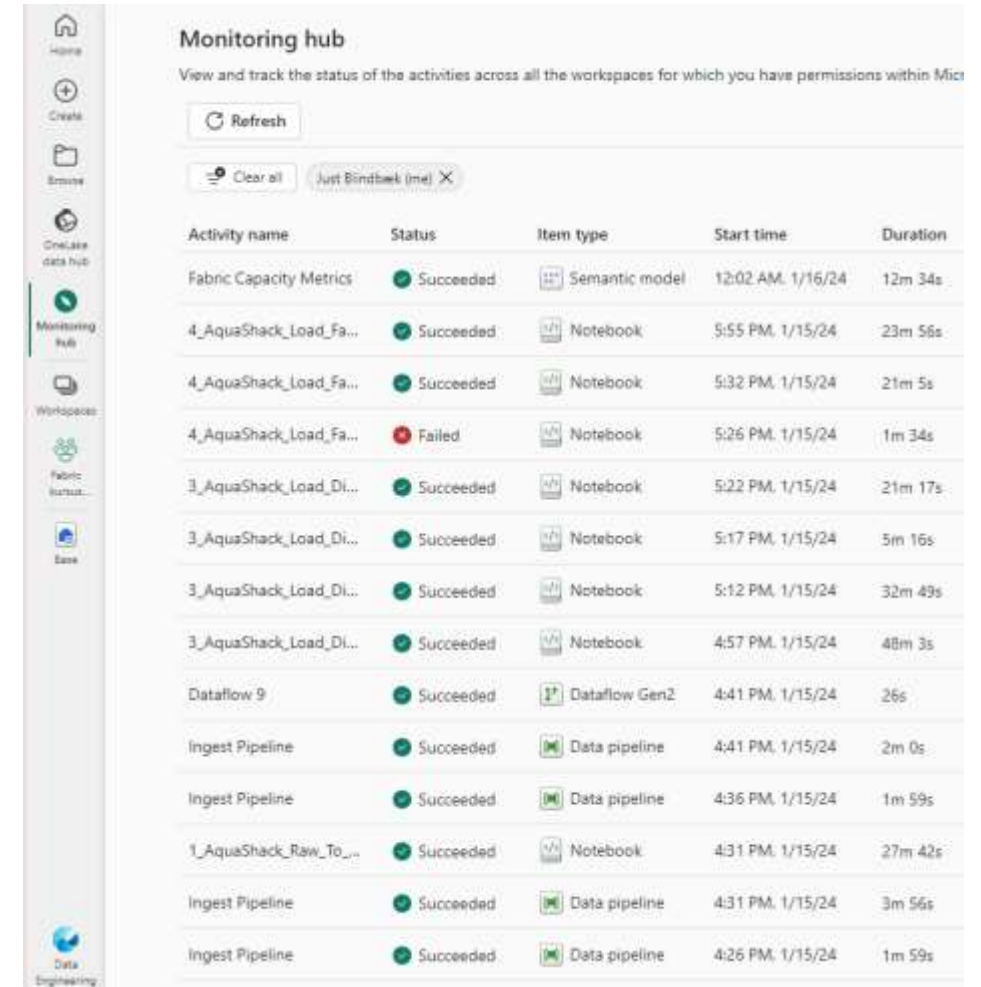


Monitoring (hub)

Quick operational overview for items

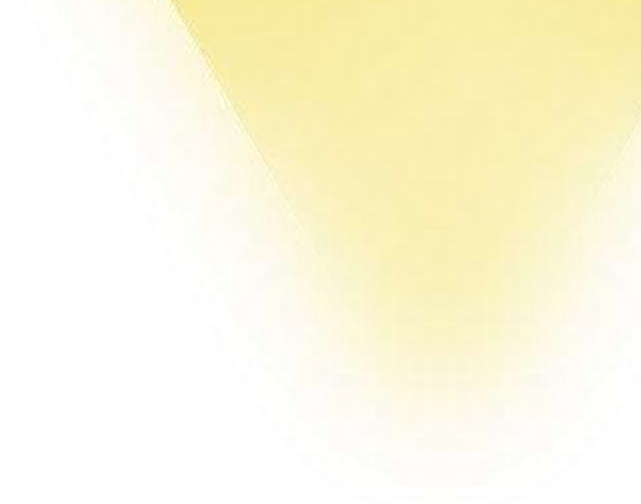
Monitoring (hub): Quick operational overview for items

- Monitor various Microsoft Fabric activities, such as semantic model refresh and Spark Job runs
- Is available for workloads from Power BI, Data Factory, Data Engineering and Data Science
- Filters are applied by default to limit the number of items initially displayed
- Select different filters by using the filter drop-down in the upper right corner
- Select an item from the list and get detailed information about that item
- Hover over an item's name and get available quick actions for the item, such as stop, start, re-run, or other quick actions



Activity name	Status	Item type	Start time	Duration
Fabric Capacity Metrics	Succeeded	Semantic model	12:02 AM, 1/16/24	12m 34s
4_AquaShack_Load_Fa...	Succeeded	Notebook	5:55 PM, 1/15/24	23m 56s
4_AquaShack_Load_Fa...	Succeeded	Notebook	5:32 PM, 1/15/24	21m 5s
4_AquaShack_Load_Fa...	Failed	Notebook	5:26 PM, 1/15/24	1m 34s
3_AquaShack_Load_Di...	Succeeded	Notebook	5:22 PM, 1/15/24	21m 17s
3_AquaShack_Load_Di...	Succeeded	Notebook	5:17 PM, 1/15/24	5m 16s
3_AquaShack_Load_Di...	Succeeded	Notebook	5:12 PM, 1/15/24	32m 49s
3_AquaShack_Load_Di...	Succeeded	Notebook	4:57 PM, 1/15/24	48m 3s
Dataflow 9	Succeeded	Dataflow Gen2	4:41 PM, 1/15/24	26s
Ingest Pipeline	Succeeded	Data pipeline	4:41 PM, 1/15/24	2m 0s
Ingest Pipeline	Succeeded	Data pipeline	4:36 PM, 1/15/24	1m 59s
1_AquaShack_Raw_To...	Succeeded	Notebook	4:31 PM, 1/15/24	27m 42s
Ingest Pipeline	Succeeded	Data pipeline	4:31 PM, 1/15/24	3m 56s
Ingest Pipeline	Succeeded	Data pipeline	4:26 PM, 1/15/24	1m 59s

<https://learn.microsoft.com/en-us/fabric/admin/monitoring-hub>



Workspace Monitoring

Root cause analysis with diagnostic logs

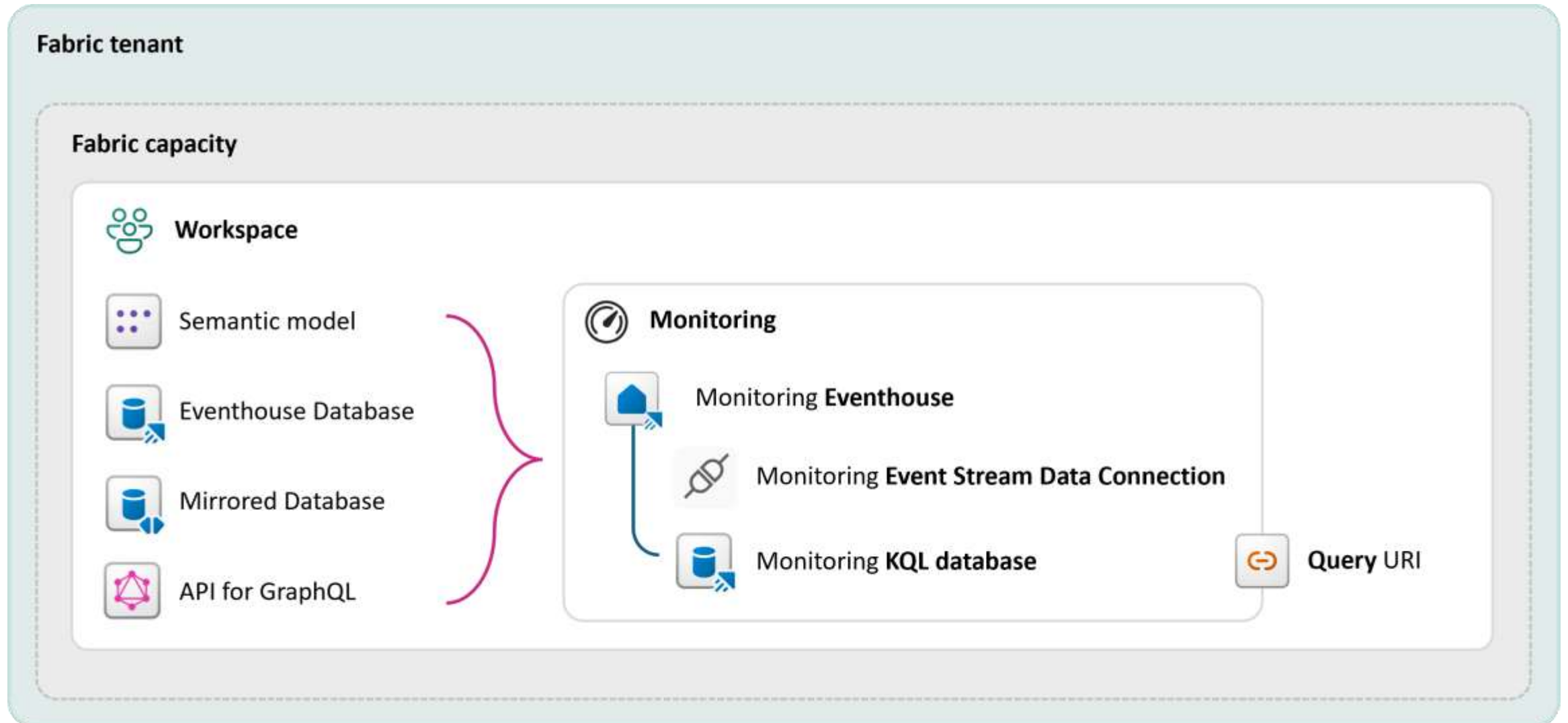
Workspace Monitoring: Root cause analysis with diagnostic logs

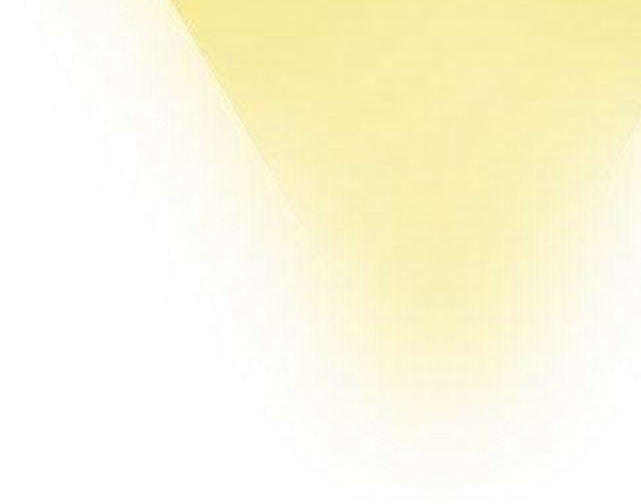
- **Built-in solution** for root-cause analysis, historical log analysis, and anomaly detection.
- Simplifies access to diagnostic logs and metrics.
- Leverages Eventhouse (KQL database) for efficient, self-serve troubleshooting and diagnostics.
- Provides seamless, end-to-end visibility across Fabric native data apps.
- Correlates events from origination to subsequent operations and queries across services.
- **Workspace-level enablement:**
 - Send diagnostics to an Eventhouse for analysis.
 - Access pre-built reports, dashboards and Querysets for quick insights.
 - Raw data available for custom dashboards, insights, and alerts.



<https://learn.microsoft.com/en-us/fabric/fundamentals/workspace-monitoring-overview> and
<https://github.com/microsoft/fabric-toolbox/tree/main/monitoring/workspace-monitoring-dashboards>

Workspace Monitoring: Scope



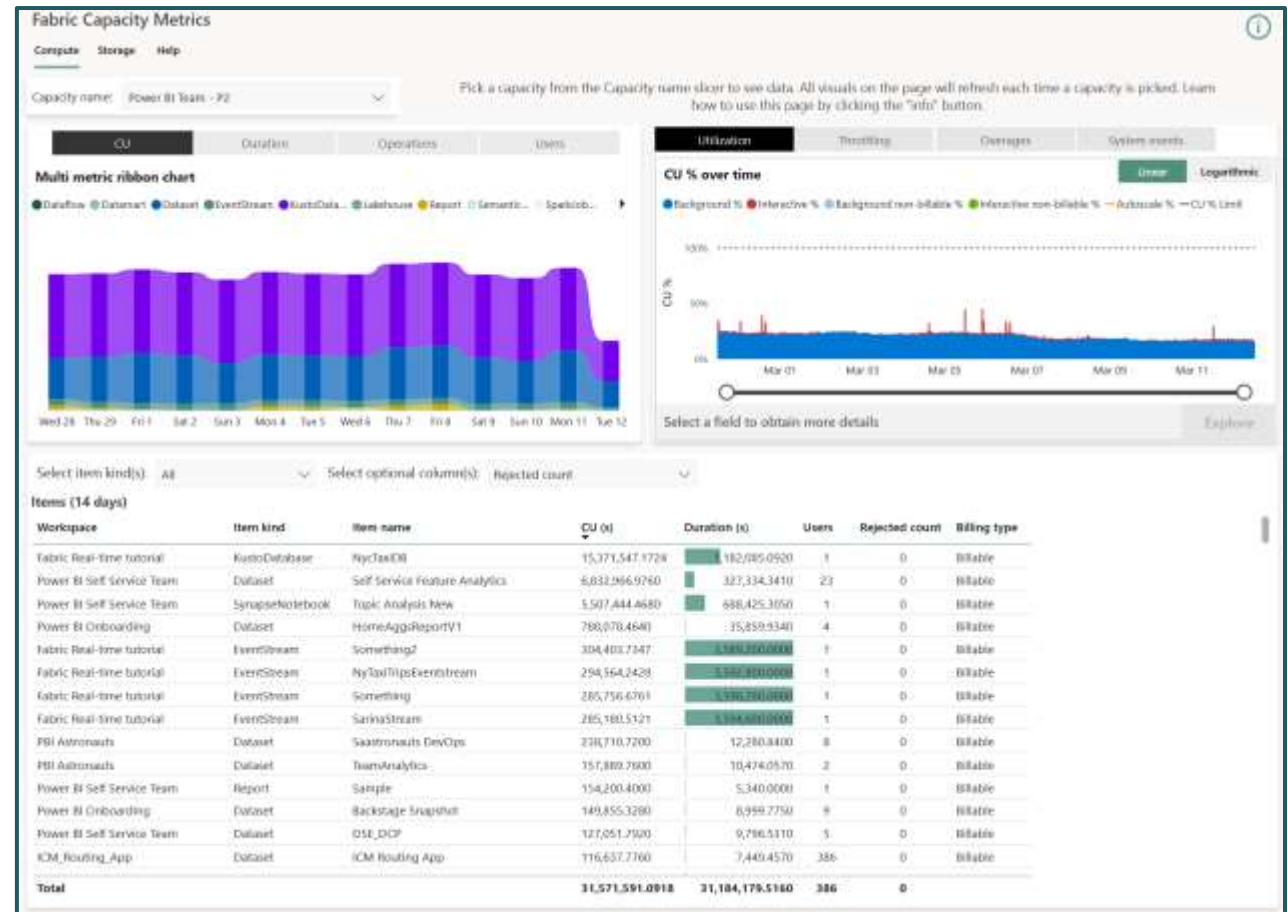


Capacity Metrics App

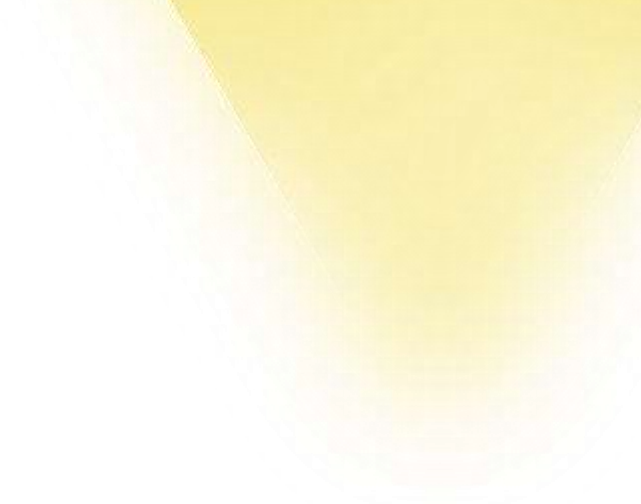
Sizing & performance optimization

Capacity Metrics App: Sizing & performance optimization

- Gain tenant-wide visibility into capacity usage across all Fabric workloads.
- Identify resource usage trends to optimize autoscale and mitigate throttling impacts.
- Compare preview and production workloads for data-driven capacity sizing decisions.
- Zoom in on workload operations and artifacts with detailed granularity down to 30 seconds.
- Monitor real-time operations and the impact of long-running jobs on capacity limits.
- Analyze user experience to plan efficient scale-ups and optimize resource allocation.



<https://learn.microsoft.com/en-us/fabric/enterprise/metrics-app>



Admin Monitoring Workspace

Insights in One Place

Admin Monitoring Workspace: Insights in One Place

- Tenant-wide governance & adoption insights
- One-stop shop for enterprise reporting and analytics
- Data queried from multiple sources, transformed, and landed in a single repository specific to each tenant
- Includes out of the box reporting focused on Fabric tenant management scenarios
- Also includes semantic models for customization
 - Out of the box reporting and curated semantic models made readily available via the workspace
 - Managed, automated data refresh of all semantic models
- Available to all tenants regardless of licensing type or total # of users
- Only retains information for 28 day

	Name
	Content Sharing
	Content Sharing
	Feature Usage and Adoption
	Feature Usage and Adoption
	Purview Hub
	Purview Hub

<https://learn.microsoft.com/en-us/fabric/admin/monitoring-workspace>

Feature usage and adoption report

- Leverages audit data (previously only available via the Activity Events API) for conducting audits and understanding how various Fabric features are utilized across your tenant.
- Audit and inventory-focused
 - Understand what activities are occurring in your tenant, by whom, on which item, and where
 - Audit combined with tenant inventory for understanding your most heavily-utilized and 'dormant' items
 - Analyze custom scenarios via drill through pages and flexible visuals (e.g. decomp tree)



<https://learn.microsoft.com/en-us/fabric/admin/feature-usage-adoption>

Content sharing report

- Understand how Fabric items are distributed and shared across the organization:
 - Is my organization effectively using domains to organize its Fabric inventory?
 - Most shared items by total access count
 - Item distribution by endorsement and sensitivity label
 - Items shared with the entire organization using links
 - Item deleted within the last 28 days



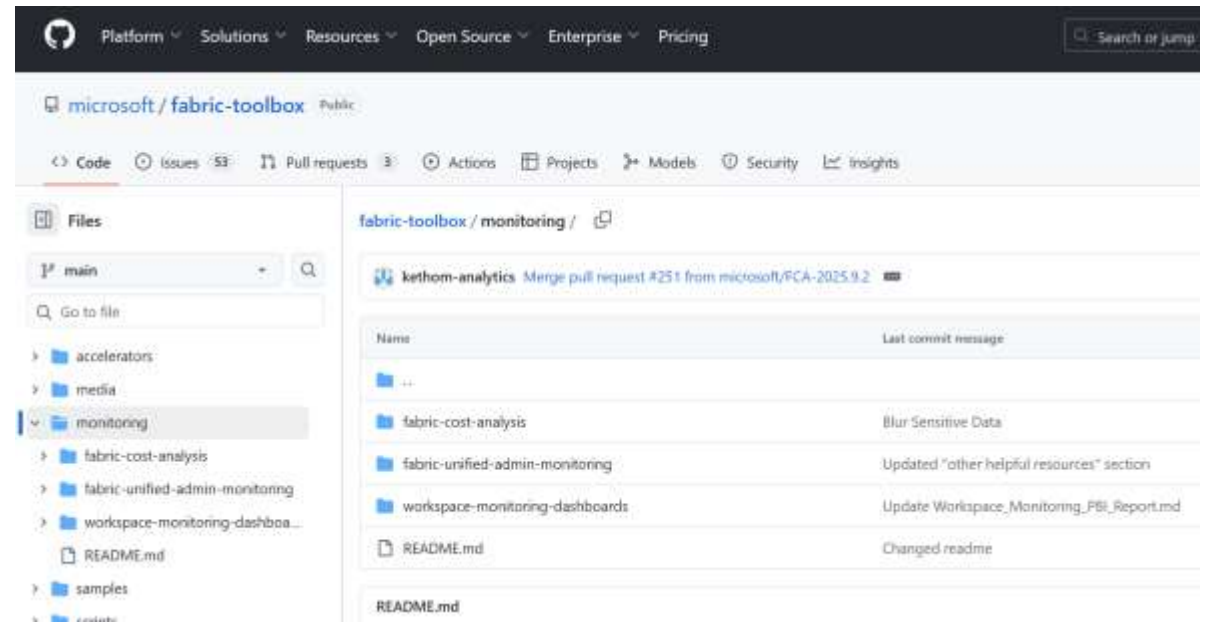
<https://learn.microsoft.com/en-us/fabric/admin/content-sharing>

Community Tools

1. Fabric Unified Admin
Monitoring (FUAM)

Fabric Toolbox

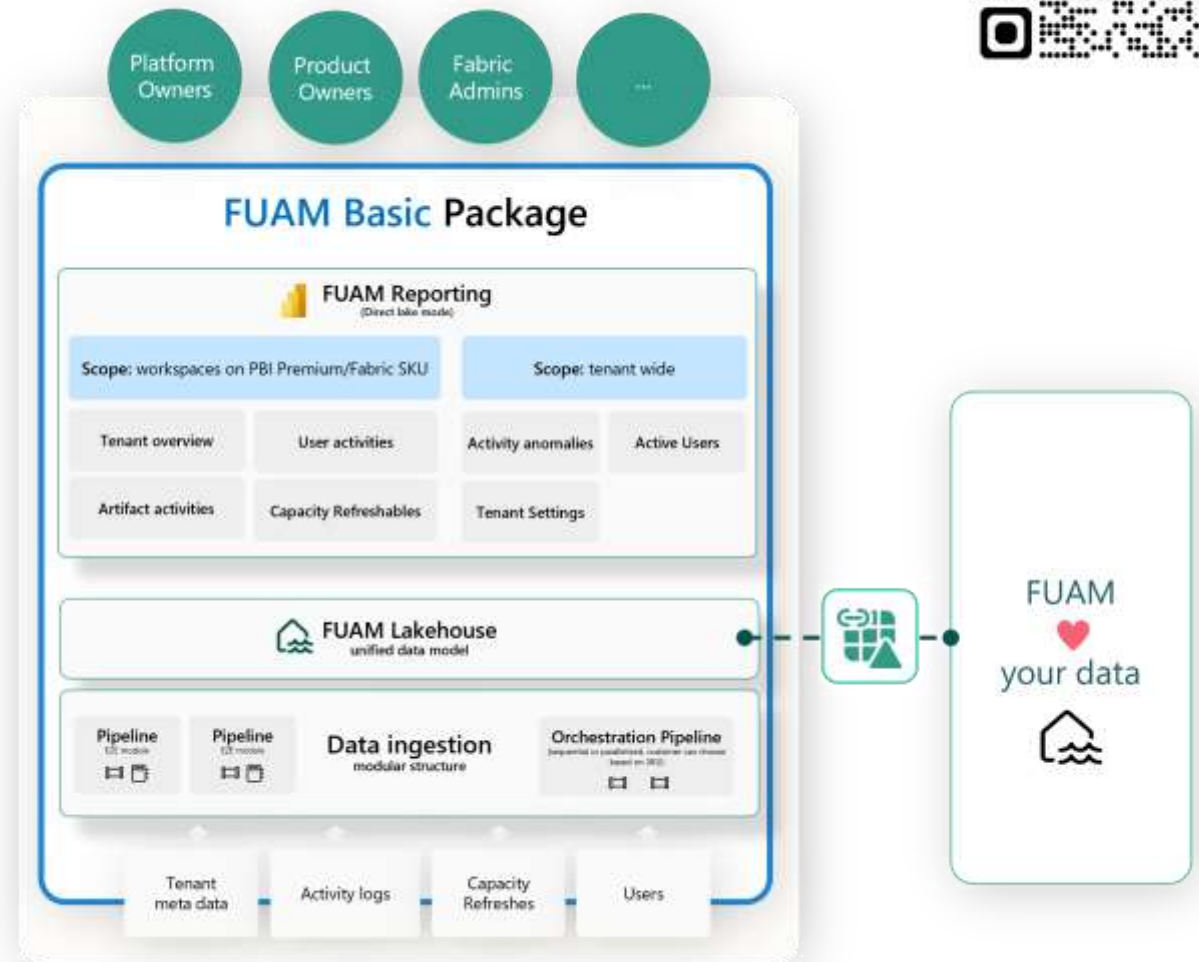
- Set of tools created by Microsoft employees but as part of the community
- On GitHub
- Some tools for monitoring
 - Fabric Unified Admin Monitoring
 - Workspace Monitoring Dashboards
 - Fabric Cost Analysis



Fabric Unified Admin Monitoring (FUAM)



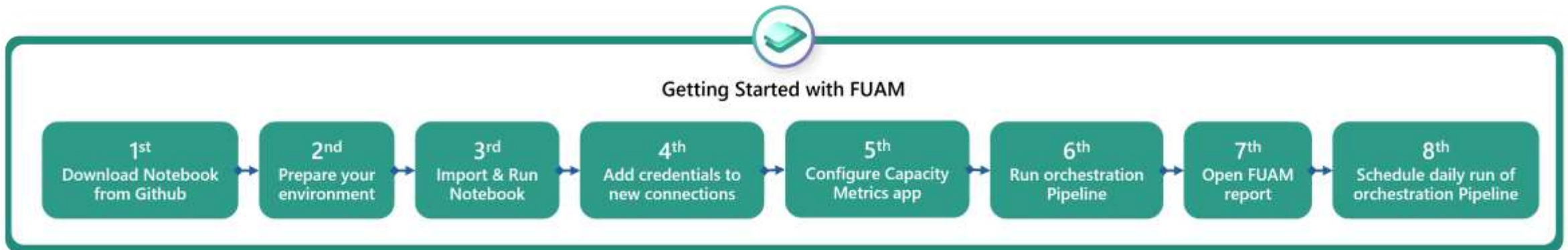
- Holistic, community-driven monitoring across all levels.
- A solution designed to deliver holistic monitoring across Fabric.
- Fully built using Microsoft Fabric tools, leveraging Data Pipelines and Notebooks to extract and transform monitoring data.
- Provides a unified view of various telemetry sources, enabling both high-level overviews and granular deep dives into specific artifacts.



Fabric Unified Admin Monitoring (FUAM)



- Supports monitoring of:
 - Tenant Settings, Delegated Tenant Settings, Activities, Workspaces, Capacities, Tenant meta data (Scanner API), Capacity Refreshables and Git Connection
- Set up FUAM step by step via Notebook.



<https://github.com/microsoft/fabric-toolbox/tree/main/monitoring/fabric-unified-admin-monitoring>

Build your own solution

1. Tenant Monitoring
2. Job Monitoring
3. Capacity Monitoring

Tenant Monitoring

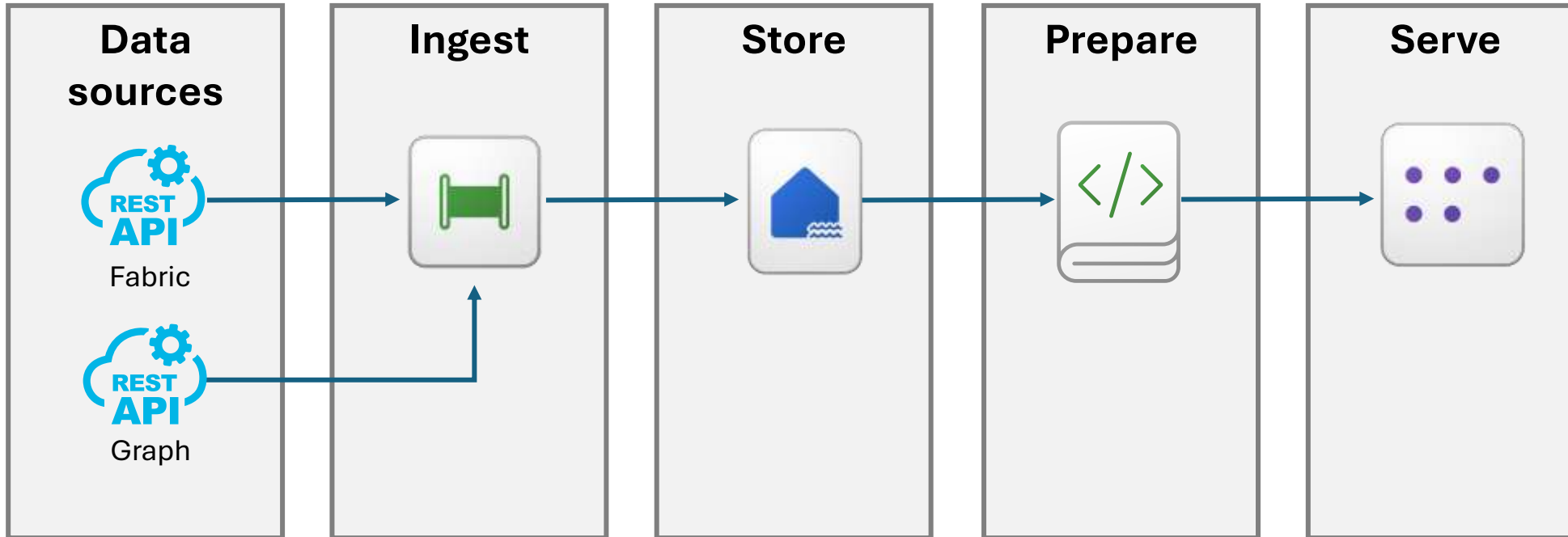
Build your own solution

A complete solution to ingest and prepare

- Fabric activity/audit events
- Fabric artifacts metadata – the tenant catalogue
- Fabric tenant settings
- Extra: Microsoft Graph data (users, groups, licenses)
- Coming soon: Fabric Pipeline templates to download and get you started

<https://github.com/justBlindbaek/PowerBIMonitor>

Dataflow architecture



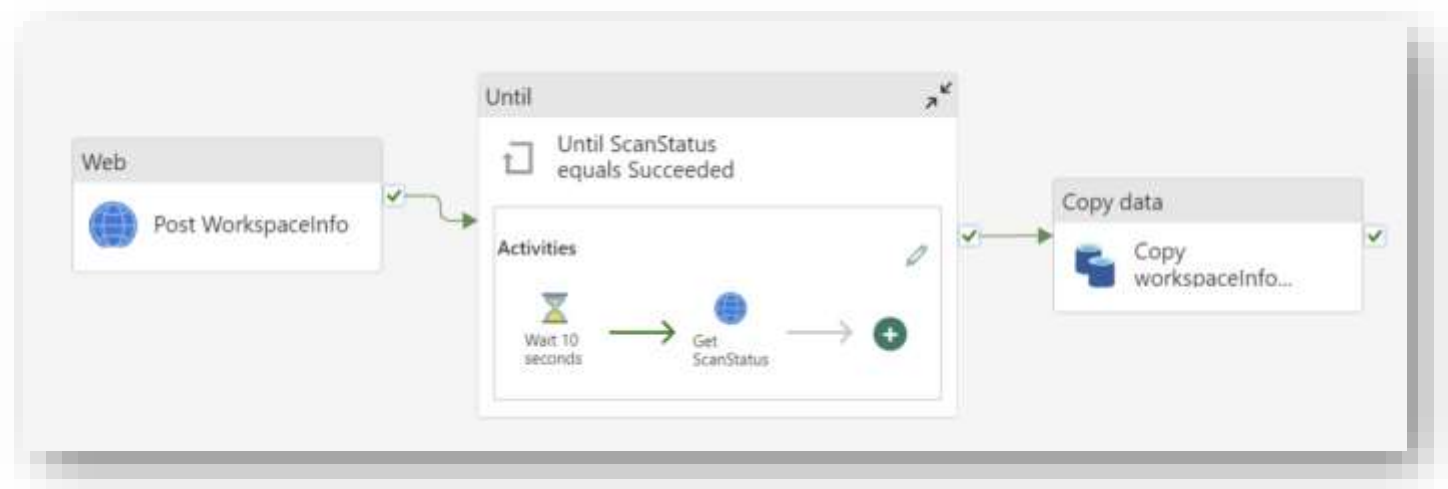
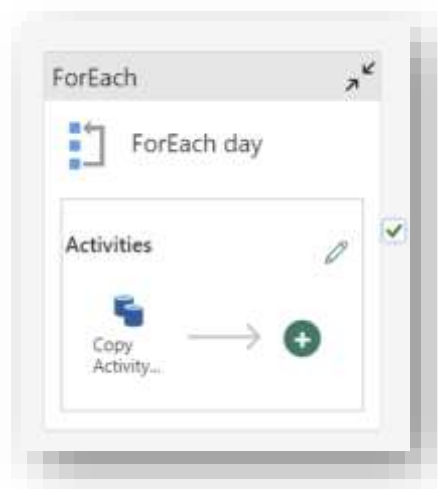
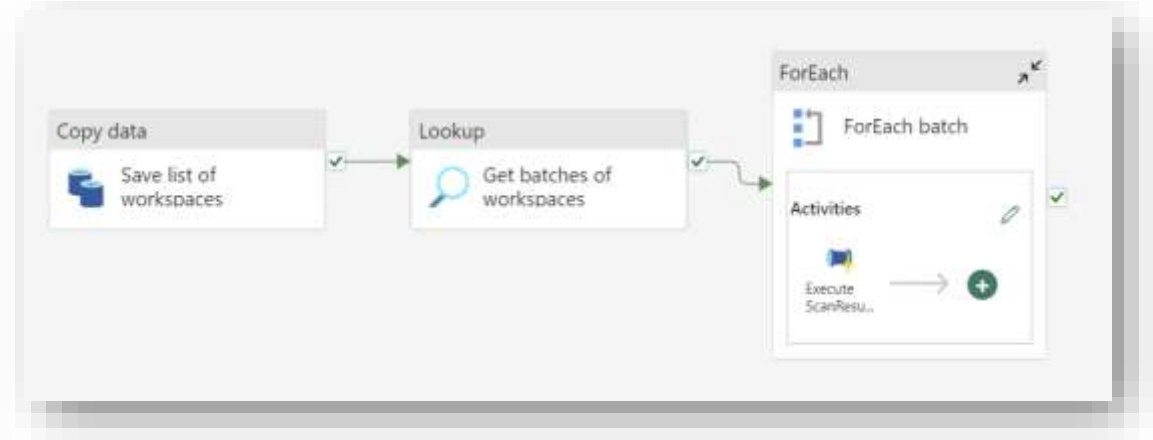
Access to the data: pre-req setup

- Create a Service Principal (App Registration)
- Create a new Security Group in Azure Active Directory
- Add the Service Principal as a member of the security group
- Enable service principal authentication for read-only admin APIs in Admin Tenant Settings and add the security group
- Optional: Enable the Enhance admin APIs responses with “detailed metadata” and ”DAX and mashup expressions”

<https://justb.dk/blog/2021/02/extracting-the-power-bi-activity-log-with-data-factory/>

Ingest with Data Factory Pipelines

	Name	Type
	PL_FabricGetTenantSettings	Data pipeline
	PL_PowerBIGetActivityEvents	Data pipeline
	PL_PowerBIWorkspaceInfo	Data pipeline
	PL_PowerBIWorkspaceInfoScanResult	Data pipeline

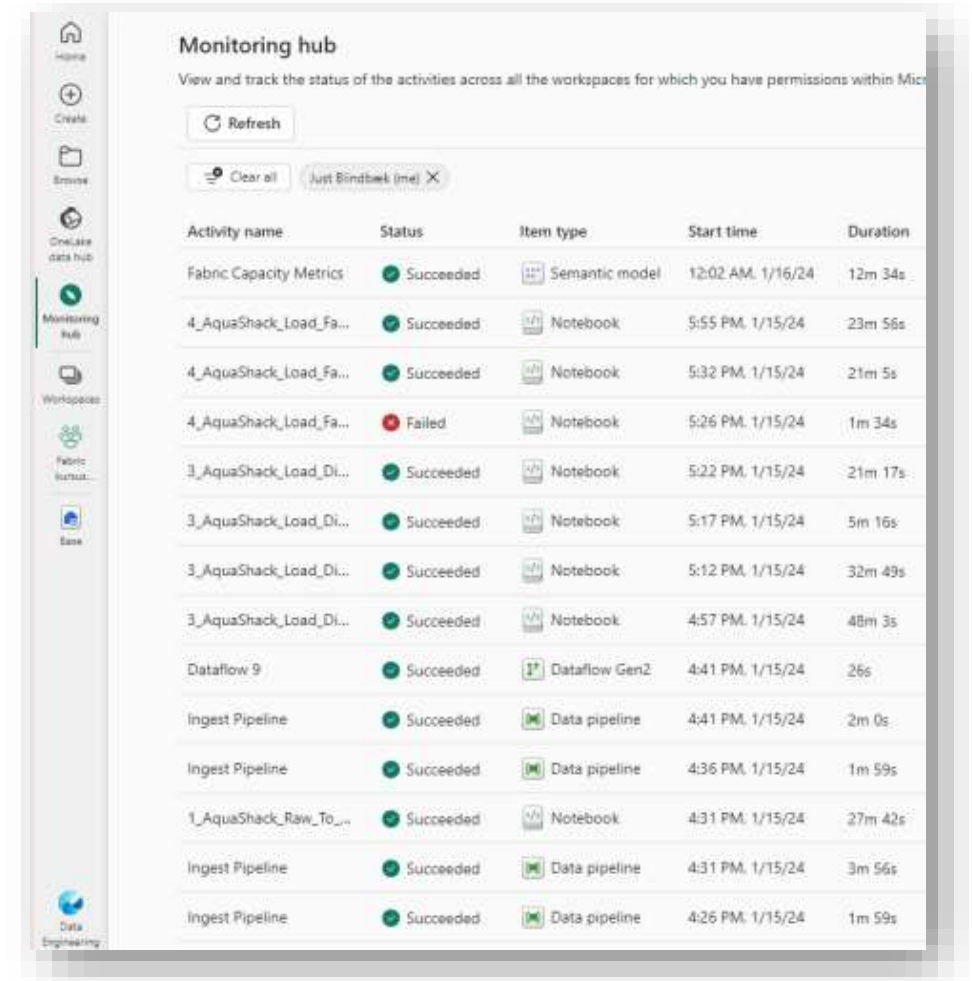


Job Monitoring

Build your own solution

Job Monitoring

- Built-in monitoring hub is fine but also very limited
 - No reports or dashboards
 - No pro-active alerts
- But then what?
 - Use the unsupported API behind the Job Monitoring list?



The screenshot shows the 'Monitoring hub' interface in Azure. It features a sidebar with navigation options: Home, Create, Browse, OnLake data hub, Monitoring hub (selected), Workspaces, Fabric instant..., and Base. The main content area displays a table of activities. At the top, there are buttons for 'Refresh', 'Clear all', and a user selection dropdown 'Just Bindbak (me) X'. The table has five columns: Activity name, Status, Item type, Start time, and Duration. The activities listed include 'Fabric Capacity Metrics', several '4_AquaShack_Load_Fa...' jobs (some succeeded, one failed), '3_AquaShack_Load_Di...' jobs, 'Dataflow 9', and multiple 'Ingest Pipeline' jobs.

Activity name	Status	Item type	Start time	Duration
Fabric Capacity Metrics	Succeeded	Semantic model	12:02 AM, 1/16/24	12m 34s
4_AquaShack_Load_Fa...	Succeeded	Notebook	5:55 PM, 1/15/24	23m 56s
4_AquaShack_Load_Fa...	Succeeded	Notebook	5:32 PM, 1/15/24	21m 5s
4_AquaShack_Load_Fa...	Failed	Notebook	5:26 PM, 1/15/24	1m 34s
3_AquaShack_Load_Di...	Succeeded	Notebook	5:22 PM, 1/15/24	21m 17s
3_AquaShack_Load_Di...	Succeeded	Notebook	5:17 PM, 1/15/24	5m 16s
3_AquaShack_Load_Di...	Succeeded	Notebook	5:12 PM, 1/15/24	32m 49s
3_AquaShack_Load_Di...	Succeeded	Notebook	4:57 PM, 1/15/24	48m 3s
Dataflow 9	Succeeded	Dataflow Gen2	4:41 PM, 1/15/24	26s
Ingest Pipeline	Succeeded	Data pipeline	4:41 PM, 1/15/24	2m 0s
Ingest Pipeline	Succeeded	Data pipeline	4:36 PM, 1/15/24	1m 59s
1_AquaShack_Raw_To...	Succeeded	Notebook	4:31 PM, 1/15/24	27m 42s
Ingest Pipeline	Succeeded	Data pipeline	4:31 PM, 1/15/24	3m 56s
Ingest Pipeline	Succeeded	Data pipeline	4:26 PM, 1/15/24	1m 59s

Job Scheduler API

- Fabric REST API: List Item Job Instances
- Returns a list of “job instances” for a specified item

```
{
  "value": [
    {
      "id": "f2d65699-dd22-4889-980c-15226deb0e1b",
      "itemId": "431e8d7b-4a95-4c02-8ccd-6faef5ba1bd7",
      "jobType": "DefaultJob",
      "invokeType": "Manual",
      "status": "Completed",
      "rootActivityId": "8c2ee553-53a4-7edb-1042-0d8189a9e0ca",
      "startTimeUtc": "2024-06-22T06:35:00.7812154",
      "endTimeUtc": "2024-06-22T06:35:00.8033333",
      "failureReason": null
    },
    {
      "id": "c0c99aed-be56-4fe0-a6e5-6de5fe277f16",
      "itemId": "431e8d7b-4a95-4c02-8ccd-6faef5ba1bd7",
      "jobType": "DefaultJob",
      "invokeType": "Manual",
      "status": "Completed",
      "rootActivityId": "c0c99aed-be56-4fe0-a6e5-6de5fe277f16",
      "startTimeUtc": "2024-06-22T06:35:00.7812154",
      "endTimeUtc": "2024-06-22T07:35:00.8033333",
      "failureReason": null
    }
  ]
}
```

Job Events in Real-Time Hub

- Subscribe to changes produced when Fabric runs a job
 - React to changes when refreshing a semantic model, running a scheduled pipeline, or running a notebook.
 - Each of these activities can generate a corresponding job, which in turn generates a set of corresponding job events.
- Monitor job results in time and set up alerts using Data Activator alerting capabilities
 - When the scheduler triggers a new job, or a job fails, you can receive an email alert.
 - Even if you aren't in front of the computer, you can still get the information you care about.

<https://learn.microsoft.com/en-us/fabric/real-time-hub/explore-fabric-job-events>

Capacity Monitoring

Build your own solution

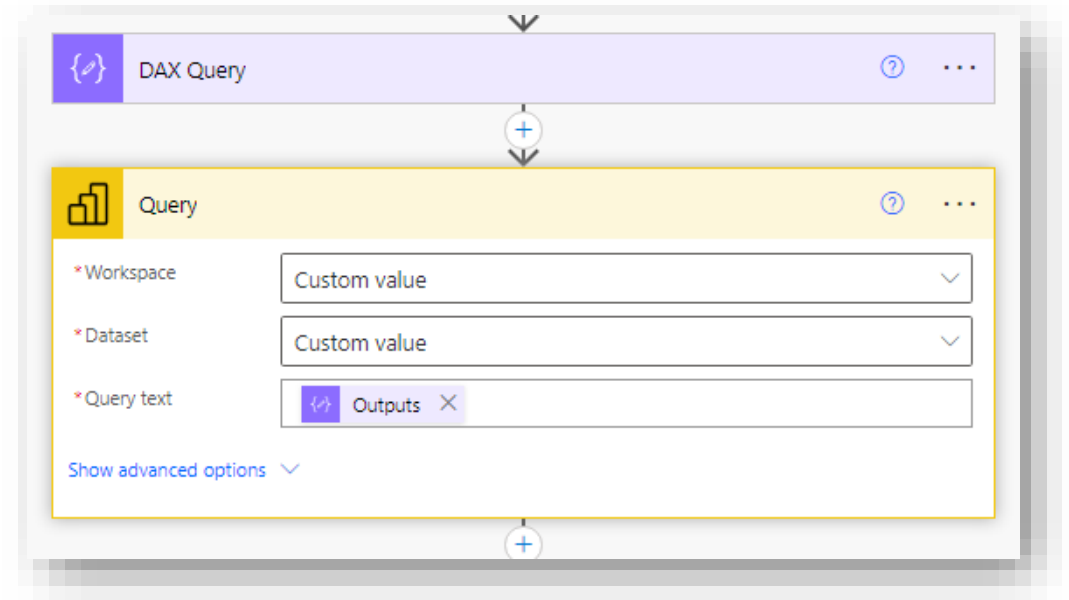
Modify the Capacity Metrics App to meet your needs

- Build a custom report off the semantic model
- Send DAX queries to the metrics app semantic model in your own solution
 - Power Automate, Notebook (SemPy), PowerShell, etc.
 - Get throttling % values (Interactive Delay, Interactive Rejection, and/or Background Rejection)
 - Latest values and/or trends over time
 - Best for summarized data only (e.g., hour, day)
- Incorporate Metrics App queries into custom solutions

```
8 # Get max date from current delta table (to avoid loading duplicate days)
9 try:
10     df_max = spark.sql(f'''
11         SELECT MAX(Date) as MaxDate
12         FROM throttling;
13     ''')
14 )
15 maxdate = df_max.first()['MaxDate']
16 except:
17     maxdate = datetime.today() + timedelta(days=-6)
18 maxdateforDAX = maxdate.strftime('%Y,%m,%d')
19
20 if maxdate.date() < (datetime.today() + timedelta(days=-1)).date():
21
22     # Get data for each capacity, write daily csv and append delta
23     for capacity in lst_capacities:
24         querytext = '''\
25             DEFINE
26             MPARAMETER 'CapacityID' = "{capID}"
27             VAR yesterday =
28                 FILTER(ALL('Dates'[Date] ), 'Dates'[Date] < TODAY() && 'Dates'[Date] > DATE({MD}) )
29
30             EVALUATE
31             SUMMARIZECOLUMNS(
32                 'Dates'[Date],
33                 'TimePoints'[Start of Hour],
34                 yesterday,
35                 "IntDelay", ROUND( 'All Measures'[Dynamic InteractiveDelay %] * 100, 2 ),
36                 "IntReject", ROUND( 'All Measures'[Dynamic InteractiveRejection %] * 100, 2 ),
37                 "BackReject", ROUND( 'All Measures'[Dynamic BackgroundRejection %] * 100, 2 )
38             )
39             '''.format(capID=capacity, MD=maxdateforDAX)
40     df_throttling = fabric.evaluate_dax(workspace=MetricsWS, dataset=MetricsModel, dax_string=querytext)
41     if len(df_throttling) >= 1:
42         df_throttling.columns = df_throttling.columns.str.replace(r'(.*)\[(\)](.*)', '', regex=True)
43         df_throttling.columns = df_throttling.columns.str.replace(' ', '_')
44         df_throttling['capacityId'] = capacity
45         filename = capacity + '_throttling_' + (datetime.today()).strftime('%Y%m%d') + '.csv'
46         df_throttling.to_csv("/lakehouse/default/Files/ThrottlingData/" + filename)
47         spk_throttle = spark.createDataFrame(df_throttling)
48         spk_throttle.write.mode("append").format("delta").option("overwriteSchema", "true").saveAsTable('Throttling')
```

Modify the Capacity Metrics App to meet your needs

- Build a custom report off the semantic model
- Send DAX queries to the metrics app semantic model in your own solution
 - Power Automate, Notebook (SemPy), PowerShell, etc.
 - Get throttling % values (Interactive Delay, Interactive Rejection, and/or Background Rejection)
 - Latest values and/or trends over time
 - Best for summarized data only (e.g., hour, day)
- Incorporate Metrics App queries into custom solutions



Fabric Capacity Insights

by DataPulse

This report provides key information on your tenant Fabric: which workspace is consuming too much power? Is my capacity properly sized? With precise graphs and tables, you'll find all the information you need to effectively manage your tenant Fabric.

Select your period

Last 30 Days

9/4/2024 - 10/3/2024

+ CONSUMPTION SUMMARY

+ CONSUMPTION ALERTS

Capacity	SKU	Capacity utilisation	Hourly overruns	Storage (GB)
Marketing capacity	FT1	43 %	25	174K
HR capacity	F2	200 %	1	0K
IT capacity	FT1	21 %	1	134K
Finance capacity	FT1	11 %	0	2K
Global		25 %	27	310K



4

Active capacities



33

Workspaces

Including 38 datasets



31

Lakehouses



40

Warehouses



7

Dataflows



8

Pipelines



22

Notebooks

Last update : 03/10/2024 at 9h

Fabric Monitoring – Feature Comparison

Feature / Tool	Monitoring Hub	Admin Monitoring Workspace	Capacity Metrics App	Workspace Monitoring	Fabric Unified Admin Monitoring
Scope	Operational monitoring of activities (e.g., semantic model refresh, Spark jobs)	Tenant-level overview, adoption, inventory, auditing	Tenant-wide capacity usage and trends	Workspace-level diagnostic logs, root cause analysis	Tenant-wide holistic monitoring (activities, capacities, workspaces, settings)
Target Users	Admins & Developers needing quick status checks	Global/Fabric/Capacity admins, COE leads	Capacity admins, Fabric admins	Developers, support teams, workspace admins	Advanced admins, COEs needing extendable monitoring
Data Sources	Activity logs from workloads (Power BI, Data Factory, Spark, Data Science)	Multiple telemetry sources, audit logs, activity events	Capacity telemetry from Fabric workloads	Eventhouse (KQL-based diagnostic logs)	Activity logs, API data (Scanner API, Admin APIs)
Retention Period	Dynamic, based on activity	28 days	12 days	Historical logs stored in Eventhouse	Depends on implementation (can be extended)
Customization	Limited filters & views	Build custom reports off semantic models	Build custom reports off semantic model	Raw data available for custom dashboards	Fully customizable (open-source, extensible via Fabric tools)
Proactive Alerting	None (manual monitoring)	None out of the box	None out of the box	Possible via KQL & integrations	None out of the box
Ease of Use	Simple UI, quick actions, minimal setup	Managed workspace, automated refresh	Visual dashboards, interactive metrics	Requires setup, KQL skills useful	Requires setup with Data Pipelines & Notebooks
Availability	Built-in, available to all Fabric tenants	Available to all tenants regardless of license	Built-in, Fabric capacity-level	Requires Eventhouse (uses capacity)	Community-supported, uses Fabric components

Fabric Monitoring – When to Use What

Scenario / Need	Monitoring Hub	Admin Monitoring Workspace	Capacity Metrics App	Workspace Monitoring	Fabric Unified Admin Monitoring
Daily operational monitoring (refreshes, jobs, failures)	✅ Simple & quick for recent activity	❌ Not real-time	❌ Focused on capacity, not ops	✅ Deep dive into logs & anomalies	❌ Dont include job events
Tenant-wide adoption, inventory, governance insights	❌ Too limited	✅ Best choice for overview & reports	❌ Not focused on adoption	❌ Workspace-specific only	✅ Combines adoption, inventory, settings
Capacity usage & optimization	❌ No capacity view	❌ Not designed for capacity	✅ Primary tool for capacity monitoring	⚠️ Workspace-level capacity insight	✅ Custom capacity dashboards possible
Root cause analysis & diagnostics (troubleshooting)	❌ High-level only	❌ Not designed for RCA	❌ Focused on trends, not diagnostics	✅ Eventhouse-powered deep analysis	✅ Custom diagnostics possible via Notebooks
Custom dashboards, flexible reporting, extendable solution	❌ Minimal customization	✅ Custom visuals via semantic models	✅ Custom visuals via semantic model	✅ Build custom KQL queries	✅ Full flexibility to build custom reports & alerting
Proactive alerting & automation	❌ No alerting	❌ No alerting	❌ Needs custom integration	⚠️ Possible with KQL + integrations	❌ No alerting
Low maintenance, out-of-the-box experience	✅ Instant availability	✅ Managed & automated refresh	✅ Built-in app, minimal setup	⚠️ Needs setup on workspace level	❌ Requires deployment & maintenance (code first)
Best suited for...	Daily ops & quick insights	Tenant governance & audits	Capacity admins & performance planners	Troubleshooting & support teams	COEs & advanced admins needing tailored monitoring

Fabric Monitoring: Quick Rule of Thumb

- **Monitoring Hub** = “What’s going on now?” (Ops dashboard)
- **Admin Monitoring Workspace** = “What’s happening across my tenant?” (Governance & adoption)
- **Capacity Metrics App** = “Are we running out of resources?” (Capacity & sizing)
- **Workspace Monitoring** = “Why is this workspace having issues?” (Root cause & diagnostics)
- **FUAM** = “I need it all, my way” (Custom, holistic, extensible monitoring)

Thank you, partners



Session Feedback 🌟💖



https://bit.ly/dMC2025_SessionFeedback